

7. ① (i) $\forall x \in \bigcup_{k=1}^m (A_{ik} \cap B), \exists l \in \{1, 2, \dots, m\}$
 $x \in A_{il} \cap B, \therefore x \in A_{il} \wedge x \in B, \therefore x \in A \wedge x \in B$
 $\therefore x \in A \cap B. \therefore \bigcup_{k=1}^m (A_{ik} \cap B) \subseteq A \cap B$
 (ii) $\forall x \in A \cap B, x \in A \wedge x \in B. \therefore \exists l \in \{1, 2, \dots, m\},$
 $x \in A_{il}, \therefore x \in A_{il} \cap B. \therefore x \in \bigcup_{k=1}^m (A_{ik} \cap B).$
 $\therefore A \cap B \subseteq \bigcup_{k=1}^m (A_{ik} \cap B)$
 综合 (i) (ii) 有 $\bigcup_{k=1}^m (A_{ik} \cap B) = A \cap B$

② $\forall A_{ij} \cap B, A_{ik} \cap B \in \pi_2,$
 $(A_{ij} \cap B) \cap (A_{ik} \cap B) = (A_{ij} \cap A_{ik}) \cap B = \emptyset$
 综合 ① ② 得 π_2 为 $A \cap B$ 的一个划分

9. R_1, R_2 为等价关系, $\therefore R_1, R_2$ 满足 r, s, t .

① 自反: $\forall x_1 \in A, y_1 \in B, \text{有 } \langle x_1, x_1 \rangle \in R_1, \langle y_1, y_1 \rangle \in R_2.$
 $\therefore \langle \langle x_1, y_1 \rangle, \langle x_1, y_1 \rangle \rangle \in R_3. \text{ 故 } R_3 \text{ 满足自反性.}$

② 对称 $\forall \langle \langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle \rangle \in R_3,$

有 $\langle x_1, x_2 \rangle \in R_1, \langle y_1, y_2 \rangle \in R_2.$

$\therefore \langle x_2, x_1 \rangle \in R_1, \langle y_2, y_1 \rangle \in R_2$

$\therefore \langle \langle x_2, y_2 \rangle, \langle y_2, y_1 \rangle \rangle \in R_3, \therefore R_3 \text{ 满足对称性.}$

③ $\forall \langle \langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle \rangle, \langle \langle x_2, y_2 \rangle, \langle x_3, y_3 \rangle \rangle \in R_3,$

$\therefore \langle x_1, x_2 \rangle, \langle x_2, x_3 \rangle \in R_1, \langle y_1, y_2 \rangle, \langle y_2, y_3 \rangle \in R_2$

$\therefore \langle x_1, x_3 \rangle \in R_1, \langle y_1, y_3 \rangle \in R_2.$

$\therefore \langle \langle x_1, y_1 \rangle, \langle x_3, y_3 \rangle \rangle \in R_3 \therefore R_3 \text{ 满足传递性.}$

综合 ① ② ③ 得 R_3 是 $A \times B$ 上的等价关系.

10. (4) 是, (1) (2) (3) 否

(1) $A = \{0, 1\} \pi_1 = \{\{0\}, \{1\}\}, \pi_2 = \{\{0, 1\}\}.$

$\pi_1 \cup \pi_2 = \{\{0\}, \{1\}, \{0, 1\}\}$

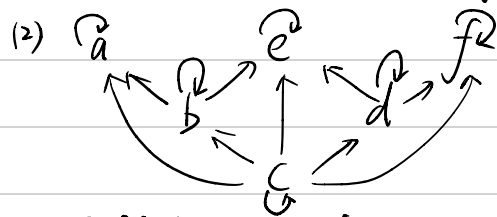
(2) $A = \{0, 1\} \pi_1 \cap \pi_2 = \emptyset.$

(3) $A = \{1, 2, 3\}, \pi_1 = \{\{1\}, \{2, 3\}\}, \pi_2 = \{\{1\}, \{2\}, \{3\}\}.$

$\pi_1 - \pi_2 = \{\{2, 3\}\}$

(4) $(\pi_1 \cap (\pi_2 - \pi_1)) \cup \pi_1 = (\pi_1 \cap \pi_2 \cap \overline{\pi_1}) \cup \pi_1 = \pi_1$

1. (1) $b \leq a, c \leq e, d \leq f, c \leq f$



(3) (a) 极大: a, e, f ; 极小: c

最小: c ; 下界: c ; 最大下界: c

(b) 极大: b, d ; 极小: b, d

上界: e ; 下界: c ; 最小上界: e ; 最大下界: c

(c) 极大/最大: e ; 极小/最小: b .

上界/最小上界: e ; 下界: b, c ; 最大下界: b .

(d) 极大/最大: e , 极小: b, d

上界/最小上界: e , 下界/最大下界: c

4. 均成立

(1) ① 自反: $\forall x \in B, \langle x, x \rangle \in B \times B, x \in A$

又 R 自反, $\therefore \langle x, x \rangle \in R. \therefore \langle x, x \rangle \in R \cap (B \times B)$

即 $\langle x, x \rangle \in R'$

② 传递: $\forall \langle a, b \rangle, \langle b, c \rangle \in R', \text{有 } \langle a, b \rangle, \langle b, c \rangle \in R$

$\therefore \langle a, c \rangle \in R. \text{ 又 } a, c \in B, \therefore \langle a, c \rangle \in B \times B.$

$\therefore \langle a, c \rangle \in R'.$

③ 反对称性 $\forall \langle a, b \rangle, \langle b, a \rangle \in R', \text{有 } \langle a, b \rangle, \langle b, a \rangle \in R.$

R 有反对称性, $\therefore a = b.$

(2) ① 自反. $\forall x \in B, x \in A, \therefore A$ 反自反, $\therefore \langle x, x \rangle \notin R.$

又 $R' = R \cap (B \times B). \therefore \langle x, x \rangle \notin R'$

② 传递性: 同 (1) ②

③ 反对称性: 同 (1) ③

(3) ① ② ③ 同 (1)

④ 完全可比性: $\forall a, b \in B, \text{有 } a, b \in A.$

$\therefore R$ 为全序, $\therefore \langle a, b \rangle \in R$ 或 $\langle b, a \rangle \in R$

又 $\langle a, b \rangle \in B \times B$ 且 $\langle b, a \rangle \in B \times B.$

$\therefore \langle a, b \rangle \in R' \text{ 或 } \langle b, a \rangle \in R'.$

(4) ①②③④同(3)

⑤ 最小元素存在性. $\forall S \subseteq B$ 且 $S \neq \emptyset$, $S \subseteq A$,
∵ A 良序, $\therefore \exists a \in S$, $\forall x \in S$, $\langle a, x \rangle \in R$.

又 $a, x \in B$, $\therefore \langle a, x \rangle \in B \times B$

$\therefore \langle a, x \rangle \in R'$.